Ruchir Bhatia | Software & ML Engineer

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Education

M.S., Computer Science, University of Southern California May 2027
- Coursework: CSCI 570: Analysis of Algorithms, CSCI 566: Deep Learning and its Applications, CSCI 642: Robot Learning

B.Tech, Computer Engineering, Bharati Vidyapeeth College of Engineering Jun 2024
- GPA: 3.87/4.00 (Top 5% of class)
- Coursework: Algorithm Design and Analysis, Operating Systems, Machine Learning

Skills

Languages: Python, C++, TypeScript/JavaScript, SQL, Swift, Bash
Backend & Distributed Systems: REST APIs, microservices, caching, real-time systems, system design, FastAPI, Node.js
Fullstack & Mobile: React, React Native, Next.js, iOS, Socket.io, Prisma ORM
Machine Learning & AI: PyTorch, TensorFlow, XGBoost/LightGBM, LLMs, RAG, Vision-Language Models, Whisper ASR
Cloud, Databases & DevOps: AWS (EC2, S3, ElastiCache), PostgreSQL, MySQL, Redis, Docker, Git, CI/CD

Experience

Transnode AI, LLC | Software Engineer Intern Jun 2026 – Aug 2026
- Owned customer-facing features for the Earnie app across React Native, iOS, and web through the full SDLC, breaking ambiguous requirements into shippable components and fixing release-blocking issues.
- Used AI-assisted development tools while validating AI-generated code, contributing to ∼111% user growth (679 to 1,436).

Bharati Vidyapeeth COE | ML Research Assistant Jan 2024 – Jun 2024
- Designed and built a RAG pipeline (Python, PyTorch, LLMs) grounding generated responses in structured knowledge via semantic retrieval, profiling each stage to find and remove bottlenecks.
- Ran controlled, reproducible experiments to validate every optimization, cutting end-to-end response latency 65%.

Redis Ltd | Software Engineer Intern Jun 2023 – Sep 2023
- Designed, built, and tested Redis 7.2 (NoSQL) caching workflows for scalable distributed systems with high-throughput, low-latency data access on AWS ElastiCache, cutting memory usage 18%.
- Automated validation and regression testing, improving testing efficiency 30% while strengthening reliability at scale.

Projects

DriveMind – Multimodal AI Driving Assistant Live Demo | github.com/RuchirBhatia4/DriveMind
- Built a real-time assistant fusing YOLO11 + ByteTrack road perception (9–18 ms/frame), MediaPipe driver monitoring, Whisper ASR, and a 4-bit vision-language model (∼2.5 s on-device voice answers).
- Designed a concurrent FastAPI/WebSocket pipeline with a worker thread per stage and latest-frame-wins backpressure, routing between on-device and cloud models with a circuit breaker for offline fallback.
- Reached 0.95 blink F1 (leave-one-person-out) with a gradient-boosted eye-state model; a pre-registered alert rule caught 10/12 unseen drowsy drivers with 1 false alert in 2 h.

PitWall AI – Live F1 Race-Strategy Engine Live App | github.com/RuchirBhatia4/Pitwall-AI
- Built and deployed a fullstack service: a FastAPI REST API on Render and a Next.js/TypeScript dashboard on Vercel, turning live F1 timing data into pit calls in under 3 s.
- Combined a Bayesian tyre-wear model, dynamic-programming search over 0–3 stop plans and Monte Carlo race simulation; walk-forward on 205 driver-races vs. a naive baseline: first-stop error 7.1 vs. 9.0 laps, stop count 53.7% vs. 44.4%.

Sociopals – Real-Time Social Platform Live App | github.com/RuchirBhatia4/sociopals
- Shipped a fullstack platform solo (Next.js, TypeScript, Socket.io, PostgreSQL/Prisma), Dockerized, live on Heroku.
- Built auth, real-time messaging, audio/video, GitHub-linked project channels, and sandboxed in-chat code execution.

Publications

IEEE Published Paper – OS Scheduling & Page Replacement Research Paper
- Peer-reviewed analysis of page-replacement and CPU scheduling algorithms for OS memory management.

Patent Accepted – LLM Healthcare Chatbot Patent
- RAG-based healthcare chatbot combining retrieval-augmented generation with structured medical knowledge for clinical QA.

Research Paper – VLM4Rec: Multimodal Recommendation Research Paper
- Grounded item images into semantic descriptions via a vision-language model, outperforming fusion-based baselines.

Honors and Awards

- 4.0 GPA & Joint Top Scorer, DSCI 552 and Information Retrieval coursework, USC M.S. in Computer Science.
- Top 5% of B.Tech Computer Engineering Class at Bharati Vidyapeeth College of Engineering.
- Registered Software Copyright for Page-Replacement Algorithms using Shell Script. (Copyright)